

SIMATS ENGINEERING

CO3- ASSESSMENT TOOL 3

ERROR ANALYSIS TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING COURSE CODE: ECA09

COURSE OUTCOME COVERED

CO 3: Evaluate finite word-length effects and multirate signal processing techniques, including decimation, interpolation, and polyphase decomposition. **Bloom Level mapped- BL3,BL4,BL5**

Topics Covered:

Quantization- Truncation and Rounding errors - Quantization noise - Limit cycle oscillations – Signal scaling. Parametric and Non parametric Spectral estimation. Decimation- Interpolation- Sampling rate conversion- Implementation of sampling rate conversion- Polyphase Decomposition - 5G Decimation/Interpolation: For mmWave beamforming

Assessment Tool number	: CO3-AT23
Assessment Tool Weightage	: 40%
Assessment Tool Name	: Error Analysis Task
Duration of this assessment	: 60 Minutes
Marks	: 25 Marks
Type of Assessment Conduction	: Hybrid

- Each student will be assigned one analytical error debugging problem as a home task.
- Students must identify the implementation error(s) and analyze the root cause.
- Perform the required calculations and provide the corrected solution.

- Justify the proposed correction with appropriate DSP concepts.
 - Upload the completed solution to Github and submit Git hub repository link in GCR.
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Dr.

Questions:

22. A DSP program computes quantization error as Input – Truncated Value, although rounding is used. Identify the programming mistake using input values [2.43, 1.89, 0.72].

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Error Identification	20%-5M				
Root Cause Analysis	30%-7.5M				
Correction	25%-6.25M				
Application of Concepts	15%-3.75M				
Clarity of Presentation	10%-2.5M				

Analytical Error Debugging Problem – Quantization Error

Problem Statement :

A DSP program computes the quantization error as:

Quantization Error = Input – Truncated Value

However, the program actually uses **rounding** instead of truncation.

Given input values:

[2.43, 1.89, 0.72]

Step 1: Identify the Implementation Error

The program uses the formula:

$$\text{Error} = \text{Input} - \text{Truncated Value}$$

But the signal is quantized using **rounding**.

Implementation Error:

- Wrong quantized value is used in the error calculation.
 - The program should subtract the **rounded value**, not the truncated value.
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Step 2: Root Cause Analysis

The programmer mixed two different quantization methods:

- **Truncation:** Removes the fractional part.
- **Rounding:** Converts to the nearest integer.

Since the DSP system performs **rounding**, using the truncated value produces incorrect quantization errors.

Step 3: Perform the Calculations

Input	Truncated Value	Rounded Value	Program Error (Input – Truncated)	Correct Error (Input – Rounded)
2.43	2	2	$2.43 - 2 = \mathbf{0.43}$	$2.43 - 2 = \mathbf{0.43}$
1.89	1	2	$1.89 - 1 = \mathbf{0.89}$	$1.89 - 2 = \mathbf{-0.11}$
0.72	0	1	$0.72 - 0 = \mathbf{0.72}$	$0.72 - 1 = \mathbf{-0.28}$

Step 4: Corrected Solution :

Since rounding is used, the correct formula is:

$$\text{Quantization Error} = \text{Input} - \text{Rounded Value}$$

Correct errors:

- $2.43 \rightarrow \mathbf{0.43}$
 - $1.89 \rightarrow \mathbf{-0.11}$
 - $0.72 \rightarrow \mathbf{-0.28}$
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Step 5: Justification Using DSP Concepts :

- In DSP, the quantization error must always be calculated with respect to the **actual quantized output**.
- If the quantizer uses **rounding**, the rounded value represents the quantized signal.
- Using the truncated value gives incorrect error values and leads to inaccurate signal analysis.
- Rounding generally provides **smaller quantization error** than truncation because it selects the nearest quantization level.

